GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 1

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 1	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.1 Safety on Raised Platforms

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify types of raised platforms used in everyday activities. Skills: Demonstrate safe practices when using or working near raised platforms. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can safety be maintained when working on raised platforms?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.1 Safety on Raised Platforms” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can safety be maintained when working on raised platforms?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to observe raised platforms in the school or community; identify hazards from pictures and realia; discuss risk-control measures; practise safe inspection and use of appropriate protective equipment under supervision.

Step 1: Learners identify hazards from safe pictures, diagrams or controlled examples and discuss who may be affected.

Step 2: In groups, learners classify risks and match them with appropriate preventive measures, warning signs and protective equipment.

Step 3: Learners practise a safety inspection or demonstrate a safe procedure using a simulation or non-hazardous equipment under supervision.

Step 4: Learners prepare a short safety checklist and explain how responsible behaviour prevents accidents.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can safety be maintained when working on raised platforms? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 2

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 1	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.1 Safety on Raised Platforms

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Explain hazards associated with working on raised platforms and ways of preventing accidents. Skills: Demonstrate safe practices when using or working near raised platforms. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can safety be maintained when working on raised platforms?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.1 Safety on Raised Platforms” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can safety be maintained when working on raised platforms?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to observe raised platforms in the school or community; identify hazards from pictures and realia; discuss risk-control measures; practise safe inspection and use of appropriate protective equipment under supervision.

Step 1: Learners identify hazards from safe pictures, diagrams or controlled examples and discuss who may be affected.

Step 2: In groups, learners classify risks and match them with appropriate preventive measures, warning signs and protective equipment.

Step 3: Learners practise a safety inspection or demonstrate a safe procedure using a simulation or non-hazardous equipment under supervision.

Step 4: Learners prepare a short safety checklist and explain how responsible behaviour prevents accidents.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can safety be maintained when working on raised platforms? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 3

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 1	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.1 Safety on Raised Platforms

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Demonstrate safe practices when using or working near raised platforms. Skills: Demonstrate safe practices when using or working near raised platforms. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can safety be maintained when working on raised platforms?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.1 Safety on Raised Platforms” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can safety be maintained when working on raised platforms?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to observe raised platforms in the school or community; identify hazards from pictures and realia; discuss risk-control measures; practise safe inspection and use of appropriate protective equipment under supervision.

Step 1: Learners identify hazards from safe pictures, diagrams or controlled examples and discuss who may be affected.

Step 2: In groups, learners classify risks and match them with appropriate preventive measures, warning signs and protective equipment.

Step 3: Learners practise a safety inspection or demonstrate a safe procedure using a simulation or non-hazardous equipment under supervision.

Step 4: Learners prepare a short safety checklist and explain how responsible behaviour prevents accidents.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can safety be maintained when working on raised platforms? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 4

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 1	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.1 Safety on Raised Platforms

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify types of raised platforms used in everyday activities. Skills: Demonstrate safe practices when using or working near raised platforms. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can safety be maintained when working on raised platforms?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.1 Safety on Raised Platforms” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can safety be maintained when working on raised platforms?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to observe raised platforms in the school or community; identify hazards from pictures and realia; discuss risk-control measures; practise safe inspection and use of appropriate protective equipment under supervision.

Step 1: Learners identify hazards from safe pictures, diagrams or controlled examples and discuss who may be affected.

Step 2: In groups, learners classify risks and match them with appropriate preventive measures, warning signs and protective equipment.

Step 3: Learners practise a safety inspection or demonstrate a safe procedure using a simulation or non-hazardous equipment under supervision.

Step 4: Learners prepare a short safety checklist and explain how responsible behaviour prevents accidents.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can safety be maintained when working on raised platforms? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 5

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 2	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.1 Safety on Raised Platforms

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Explain hazards associated with working on raised platforms and ways of preventing accidents. Skills: Demonstrate safe practices when using or working near raised platforms. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can safety be maintained when working on raised platforms?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.1 Safety on Raised Platforms” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can safety be maintained when working on raised platforms?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to observe raised platforms in the school or community; identify hazards from pictures and realia; discuss risk-control measures; practise safe inspection and use of appropriate protective equipment under supervision.

Step 1: Learners identify hazards from safe pictures, diagrams or controlled examples and discuss who may be affected.

Step 2: In groups, learners classify risks and match them with appropriate preventive measures, warning signs and protective equipment.

Step 3: Learners practise a safety inspection or demonstrate a safe procedure using a simulation or non-hazardous equipment under supervision.

Step 4: Learners prepare a short safety checklist and explain how responsible behaviour prevents accidents.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can safety be maintained when working on raised platforms? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 6

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 2	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.1 Safety on Raised Platforms

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Demonstrate safe practices when using or working near raised platforms. Skills: Demonstrate safe practices when using or working near raised platforms. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can safety be maintained when working on raised platforms?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.1 Safety on Raised Platforms” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can safety be maintained when working on raised platforms?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to observe raised platforms in the school or community; identify hazards from pictures and realia; discuss risk-control measures; practise safe inspection and use of appropriate protective equipment under supervision.

Step 1: Learners identify hazards from safe pictures, diagrams or controlled examples and discuss who may be affected.

Step 2: In groups, learners classify risks and match them with appropriate preventive measures, warning signs and protective equipment.

Step 3: Learners practise a safety inspection or demonstrate a safe procedure using a simulation or non-hazardous equipment under supervision.

Step 4: Learners prepare a short safety checklist and explain how responsible behaviour prevents accidents.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can safety be maintained when working on raised platforms? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 7

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 2	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.1 Safety on Raised Platforms

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify types of raised platforms used in everyday activities. Skills: Demonstrate safe practices when using or working near raised platforms. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can safety be maintained when working on raised platforms?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.1 Safety on Raised Platforms” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can safety be maintained when working on raised platforms?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to observe raised platforms in the school or community; identify hazards from pictures and realia; discuss risk-control measures; practise safe inspection and use of appropriate protective equipment under supervision.

Step 1: Learners identify hazards from safe pictures, diagrams or controlled examples and discuss who may be affected.

Step 2: In groups, learners classify risks and match them with appropriate preventive measures, warning signs and protective equipment.

Step 3: Learners practise a safety inspection or demonstrate a safe procedure using a simulation or non-hazardous equipment under supervision.

Step 4: Learners prepare a short safety checklist and explain how responsible behaviour prevents accidents.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can safety be maintained when working on raised platforms? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 8

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 2	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.1 Safety on Raised Platforms

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Explain hazards associated with working on raised platforms and ways of preventing accidents. Skills: Demonstrate safe practices when using or working near raised platforms. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can safety be maintained when working on raised platforms?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.1 Safety on Raised Platforms” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can safety be maintained when working on raised platforms?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to observe raised platforms in the school or community; identify hazards from pictures and realia; discuss risk-control measures; practise safe inspection and use of appropriate protective equipment under supervision.

Step 1: Learners identify hazards from safe pictures, diagrams or controlled examples and discuss who may be affected.

Step 2: In groups, learners classify risks and match them with appropriate preventive measures, warning signs and protective equipment.

Step 3: Learners practise a safety inspection or demonstrate a safe procedure using a simulation or non-hazardous equipment under supervision.

Step 4: Learners prepare a short safety checklist and explain how responsible behaviour prevents accidents.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can safety be maintained when working on raised platforms? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 9

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 3	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.2 Handling Hazardous Substances

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify common hazardous substances and their hazard symbols. Skills: Demonstrate appropriate safety precautions when dealing with hazardous substances. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can hazardous substances be handled safely?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.2 Handling Hazardous Substances” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can hazardous substances be handled safely?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to examine labelled containers and safety symbols; classify examples of hazardous substances; interpret safety instructions; discuss safe storage and disposal; practise responding to a simulated spill without touching the substance.

Step 1: Learners identify hazards from safe pictures, diagrams or controlled examples and discuss who may be affected.

Step 2: In groups, learners classify risks and match them with appropriate preventive measures, warning signs and protective equipment.

Step 3: Learners practise a safety inspection or demonstrate a safe procedure using a simulation or non-hazardous equipment under supervision.

Step 4: Learners prepare a short safety checklist and explain how responsible behaviour prevents accidents.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can hazardous substances be handled safely? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 10

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 3	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.2 Handling Hazardous Substances

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Explain safe procedures for handling, storing and disposing of hazardous substances. Skills: Demonstrate appropriate safety precautions when dealing with hazardous substances. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can hazardous substances be handled safely?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.2 Handling Hazardous Substances” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can hazardous substances be handled safely?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to examine labelled containers and safety symbols; classify examples of hazardous substances; interpret safety instructions; discuss safe storage and disposal; practise responding to a simulated spill without touching the substance.

Step 1: Learners identify hazards from safe pictures, diagrams or controlled examples and discuss who may be affected.

Step 2: In groups, learners classify risks and match them with appropriate preventive measures, warning signs and protective equipment.

Step 3: Learners practise a safety inspection or demonstrate a safe procedure using a simulation or non-hazardous equipment under supervision.

Step 4: Learners prepare a short safety checklist and explain how responsible behaviour prevents accidents.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can hazardous substances be handled safely? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 11

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 3	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.2 Handling Hazardous Substances

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Demonstrate appropriate safety precautions when dealing with hazardous substances. Skills: Demonstrate appropriate safety precautions when dealing with hazardous substances. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can hazardous substances be handled safely?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.2 Handling Hazardous Substances” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can hazardous substances be handled safely?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to examine labelled containers and safety symbols; classify examples of hazardous substances; interpret safety instructions; discuss safe storage and disposal; practise responding to a simulated spill without touching the substance.

Step 1: Learners identify hazards from safe pictures, diagrams or controlled examples and discuss who may be affected.

Step 2: In groups, learners classify risks and match them with appropriate preventive measures, warning signs and protective equipment.

Step 3: Learners practise a safety inspection or demonstrate a safe procedure using a simulation or non-hazardous equipment under supervision.

Step 4: Learners prepare a short safety checklist and explain how responsible behaviour prevents accidents.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can hazardous substances be handled safely? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 12

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 3	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.2 Handling Hazardous Substances

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify common hazardous substances and their hazard symbols. Skills: Demonstrate appropriate safety precautions when dealing with hazardous substances. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can hazardous substances be handled safely?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.2 Handling Hazardous Substances” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can hazardous substances be handled safely?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to examine labelled containers and safety symbols; classify examples of hazardous substances; interpret safety instructions; discuss safe storage and disposal; practise responding to a simulated spill without touching the substance.

Step 1: Learners identify hazards from safe pictures, diagrams or controlled examples and discuss who may be affected.

Step 2: In groups, learners classify risks and match them with appropriate preventive measures, warning signs and protective equipment.

Step 3: Learners practise a safety inspection or demonstrate a safe procedure using a simulation or non-hazardous equipment under supervision.

Step 4: Learners prepare a short safety checklist and explain how responsible behaviour prevents accidents.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can hazardous substances be handled safely? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 13

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 4	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.2 Handling Hazardous Substances

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Explain safe procedures for handling, storing and disposing of hazardous substances. Skills: Demonstrate appropriate safety precautions when dealing with hazardous substances. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can hazardous substances be handled safely?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.2 Handling Hazardous Substances” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can hazardous substances be handled safely?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to examine labelled containers and safety symbols; classify examples of hazardous substances; interpret safety instructions; discuss safe storage and disposal; practise responding to a simulated spill without touching the substance.

Step 1: Learners identify hazards from safe pictures, diagrams or controlled examples and discuss who may be affected.

Step 2: In groups, learners classify risks and match them with appropriate preventive measures, warning signs and protective equipment.

Step 3: Learners practise a safety inspection or demonstrate a safe procedure using a simulation or non-hazardous equipment under supervision.

Step 4: Learners prepare a short safety checklist and explain how responsible behaviour prevents accidents.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can hazardous substances be handled safely? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 14

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 4	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.2 Handling Hazardous Substances

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Demonstrate appropriate safety precautions when dealing with hazardous substances. Skills: Demonstrate appropriate safety precautions when dealing with hazardous substances. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can hazardous substances be handled safely?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.2 Handling Hazardous Substances” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can hazardous substances be handled safely?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to examine labelled containers and safety symbols; classify examples of hazardous substances; interpret safety instructions; discuss safe storage and disposal; practise responding to a simulated spill without touching the substance.

Step 1: Learners identify hazards from safe pictures, diagrams or controlled examples and discuss who may be affected.

Step 2: In groups, learners classify risks and match them with appropriate preventive measures, warning signs and protective equipment.

Step 3: Learners practise a safety inspection or demonstrate a safe procedure using a simulation or non-hazardous equipment under supervision.

Step 4: Learners prepare a short safety checklist and explain how responsible behaviour prevents accidents.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can hazardous substances be handled safely? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 15

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 4	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.2 Handling Hazardous Substances

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify common hazardous substances and their hazard symbols. Skills: Demonstrate appropriate safety precautions when dealing with hazardous substances. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can hazardous substances be handled safely?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.2 Handling Hazardous Substances” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can hazardous substances be handled safely?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to examine labelled containers and safety symbols; classify examples of hazardous substances; interpret safety instructions; discuss safe storage and disposal; practise responding to a simulated spill without touching the substance.

Step 1: Learners identify hazards from safe pictures, diagrams or controlled examples and discuss who may be affected.

Step 2: In groups, learners classify risks and match them with appropriate preventive measures, warning signs and protective equipment.

Step 3: Learners practise a safety inspection or demonstrate a safe procedure using a simulation or non-hazardous equipment under supervision.

Step 4: Learners prepare a short safety checklist and explain how responsible behaviour prevents accidents.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can hazardous substances be handled safely? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 16

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 4	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.2 Handling Hazardous Substances

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Explain safe procedures for handling, storing and disposing of hazardous substances. Skills: Demonstrate appropriate safety precautions when dealing with hazardous substances. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can hazardous substances be handled safely?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.2 Handling Hazardous Substances” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can hazardous substances be handled safely?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to examine labelled containers and safety symbols; classify examples of hazardous substances; interpret safety instructions; discuss safe storage and disposal; practise responding to a simulated spill without touching the substance.

Step 1: Learners identify hazards from safe pictures, diagrams or controlled examples and discuss who may be affected.

Step 2: In groups, learners classify risks and match them with appropriate preventive measures, warning signs and protective equipment.

Step 3: Learners practise a safety inspection or demonstrate a safe procedure using a simulation or non-hazardous equipment under supervision.

Step 4: Learners prepare a short safety checklist and explain how responsible behaviour prevents accidents.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can hazardous substances be handled safely? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 17

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 5	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.2 Handling Hazardous Substances

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Demonstrate appropriate safety precautions when dealing with hazardous substances. Skills: Demonstrate appropriate safety precautions when dealing with hazardous substances. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can hazardous substances be handled safely?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.2 Handling Hazardous Substances” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can hazardous substances be handled safely?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to examine labelled containers and safety symbols; classify examples of hazardous substances; interpret safety instructions; discuss safe storage and disposal; practise responding to a simulated spill without touching the substance.

Step 1: Learners identify hazards from safe pictures, diagrams or controlled examples and discuss who may be affected.

Step 2: In groups, learners classify risks and match them with appropriate preventive measures, warning signs and protective equipment.

Step 3: Learners practise a safety inspection or demonstrate a safe procedure using a simulation or non-hazardous equipment under supervision.

Step 4: Learners prepare a short safety checklist and explain how responsible behaviour prevents accidents.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can hazardous substances be handled safely? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 18

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 5	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.3 Self-Exploration and Career Development

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify personal talents, abilities and interests. Skills: Analyse ethical and unethical practices in the use of talents and make an informed career choice. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can talents and abilities guide career choices?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.3 Self-Exploration and Career Development” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can talents and abilities guide career choices?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to reflect on personal interests and strengths; discuss ways of nurturing talents; match abilities to career pathways; engage a resource person where available; prepare and present a personal career pathway profile.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can talents and abilities guide career choices? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 19

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 5	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.3 Self-Exploration and Career Development

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Relate talents and abilities to relevant career pathways. Skills: Analyse ethical and unethical practices in the use of talents and make an informed career choice. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can talents and abilities guide career choices?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.3 Self-Exploration and Career Development” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can talents and abilities guide career choices?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to reflect on personal interests and strengths; discuss ways of nurturing talents; match abilities to career pathways; engage a resource person where available; prepare and present a personal career pathway profile.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can talents and abilities guide career choices? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 20

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 5	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.3 Self-Exploration and Career Development

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Analyse ethical and unethical practices in the use of talents and make an informed career choice. Skills: Analyse ethical and unethical practices in the use of talents and make an informed career choice. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can talents and abilities guide career choices?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.3 Self-Exploration and Career Development” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can talents and abilities guide career choices?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to reflect on personal interests and strengths; discuss ways of nurturing talents; match abilities to career pathways; engage a resource person where available; prepare and present a personal career pathway profile.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can talents and abilities guide career choices? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 21

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 6	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.3 Self-Exploration and Career Development

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify personal talents, abilities and interests. Skills: Analyse ethical and unethical practices in the use of talents and make an informed career choice. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can talents and abilities guide career choices?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.3 Self-Exploration and Career Development” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can talents and abilities guide career choices?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to reflect on personal interests and strengths; discuss ways of nurturing talents; match abilities to career pathways; engage a resource person where available; prepare and present a personal career pathway profile.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can talents and abilities guide career choices? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 22

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 6	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.3 Self-Exploration and Career Development

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Relate talents and abilities to relevant career pathways. Skills: Analyse ethical and unethical practices in the use of talents and make an informed career choice. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can talents and abilities guide career choices?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.3 Self-Exploration and Career Development” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can talents and abilities guide career choices?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to reflect on personal interests and strengths; discuss ways of nurturing talents; match abilities to career pathways; engage a resource person where available; prepare and present a personal career pathway profile.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can talents and abilities guide career choices? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 23

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 6	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 1.0 Foundations of Pre-Technical Studies

Sub-strand: 1.3 Self-Exploration and Career Development

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Analyse ethical and unethical practices in the use of talents and make an informed career choice. Skills: Analyse ethical and unethical practices in the use of talents and make an informed career choice. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can talents and abilities guide career choices?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “1.3 Self-Exploration and Career Development” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can talents and abilities guide career choices?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to reflect on personal interests and strengths; discuss ways of nurturing talents; match abilities to career pathways; engage a resource person where available; prepare and present a personal career pathway profile.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can talents and abilities guide career choices? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 24

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 6	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.1 Oblique Projection

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify features and uses of oblique projection. Skills: Produce and interpret simple oblique drawings using appropriate drawing instruments. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can oblique projection communicate the form of an object?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.1 Oblique Projection” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can oblique projection communicate the form of an object?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to observe three-dimensional objects and sample drawings; identify the front face and receding axes; practise drawing basic forms; add visible details and dimensions where required; display and critique drawings using agreed criteria.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can oblique projection communicate the form of an object? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 25

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 7	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.1 Oblique Projection

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Describe the steps for drawing objects using oblique projection. Skills: Produce and interpret simple oblique drawings using appropriate drawing instruments. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can oblique projection communicate the form of an object?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.1 Oblique Projection” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can oblique projection communicate the form of an object?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to observe three-dimensional objects and sample drawings; identify the front face and receding axes; practise drawing basic forms; add visible details and dimensions where required; display and critique drawings using agreed criteria.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can oblique projection communicate the form of an object? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 26

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 7	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.1 Oblique Projection

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Produce and interpret simple oblique drawings using appropriate drawing instruments. Skills: Produce and interpret simple oblique drawings using appropriate drawing instruments. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can oblique projection communicate the form of an object?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.1 Oblique Projection” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can oblique projection communicate the form of an object?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to observe three-dimensional objects and sample drawings; identify the front face and receding axes; practise drawing basic forms; add visible details and dimensions where required; display and critique drawings using agreed criteria.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can oblique projection communicate the form of an object? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 27

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 7	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.1 Oblique Projection

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify features and uses of oblique projection. Skills: Produce and interpret simple oblique drawings using appropriate drawing instruments. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can oblique projection communicate the form of an object?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.1 Oblique Projection” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can oblique projection communicate the form of an object?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to observe three-dimensional objects and sample drawings; identify the front face and receding axes; practise drawing basic forms; add visible details and dimensions where required; display and critique drawings using agreed criteria.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can oblique projection communicate the form of an object? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 28

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 7	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.1 Oblique Projection

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Describe the steps for drawing objects using oblique projection. Skills: Produce and interpret simple oblique drawings using appropriate drawing instruments. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can oblique projection communicate the form of an object?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.1 Oblique Projection” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can oblique projection communicate the form of an object?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to observe three-dimensional objects and sample drawings; identify the front face and receding axes; practise drawing basic forms; add visible details and dimensions where required; display and critique drawings using agreed criteria.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can oblique projection communicate the form of an object? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 29

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 8	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.1 Oblique Projection

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Produce and interpret simple oblique drawings using appropriate drawing instruments. Skills: Produce and interpret simple oblique drawings using appropriate drawing instruments. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can oblique projection communicate the form of an object?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.1 Oblique Projection” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can oblique projection communicate the form of an object?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to observe three-dimensional objects and sample drawings; identify the front face and receding axes; practise drawing basic forms; add visible details and dimensions where required; display and critique drawings using agreed criteria.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can oblique projection communicate the form of an object? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 30

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 8	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.1 Oblique Projection

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify features and uses of oblique projection. Skills: Produce and interpret simple oblique drawings using appropriate drawing instruments. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can oblique projection communicate the form of an object?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.1 Oblique Projection” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can oblique projection communicate the form of an object?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to observe three-dimensional objects and sample drawings; identify the front face and receding axes; practise drawing basic forms; add visible details and dimensions where required; display and critique drawings using agreed criteria.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can oblique projection communicate the form of an object? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 31

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 8	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.1 Oblique Projection

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Describe the steps for drawing objects using oblique projection. Skills: Produce and interpret simple oblique drawings using appropriate drawing instruments. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can oblique projection communicate the form of an object?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.1 Oblique Projection” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can oblique projection communicate the form of an object?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to observe three-dimensional objects and sample drawings; identify the front face and receding axes; practise drawing basic forms; add visible details and dimensions where required; display and critique drawings using agreed criteria.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can oblique projection communicate the form of an object? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 32

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 8	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.1 Oblique Projection

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Produce and interpret simple oblique drawings using appropriate drawing instruments. Skills: Produce and interpret simple oblique drawings using appropriate drawing instruments. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can oblique projection communicate the form of an object?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.1 Oblique Projection” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can oblique projection communicate the form of an object?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to observe three-dimensional objects and sample drawings; identify the front face and receding axes; practise drawing basic forms; add visible details and dimensions where required; display and critique drawings using agreed criteria.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can oblique projection communicate the form of an object? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 33

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 9	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.1 Oblique Projection

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify features and uses of oblique projection. Skills: Produce and interpret simple oblique drawings using appropriate drawing instruments. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can oblique projection communicate the form of an object?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.1 Oblique Projection” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can oblique projection communicate the form of an object?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to observe three-dimensional objects and sample drawings; identify the front face and receding axes; practise drawing basic forms; add visible details and dimensions where required; display and critique drawings using agreed criteria.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can oblique projection communicate the form of an object? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 34

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 9	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.1 Oblique Projection

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Describe the steps for drawing objects using oblique projection. Skills: Produce and interpret simple oblique drawings using appropriate drawing instruments. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can oblique projection communicate the form of an object?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.1 Oblique Projection” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can oblique projection communicate the form of an object?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to observe three-dimensional objects and sample drawings; identify the front face and receding axes; practise drawing basic forms; add visible details and dimensions where required; display and critique drawings using agreed criteria.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can oblique projection communicate the form of an object? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 35

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 9	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.1 Oblique Projection

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Produce and interpret simple oblique drawings using appropriate drawing instruments. Skills: Produce and interpret simple oblique drawings using appropriate drawing instruments. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can oblique projection communicate the form of an object?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.1 Oblique Projection” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can oblique projection communicate the form of an object?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to observe three-dimensional objects and sample drawings; identify the front face and receding axes; practise drawing basic forms; add visible details and dimensions where required; display and critique drawings using agreed criteria.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can oblique projection communicate the form of an object? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 36

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 9	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.1 Oblique Projection

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify features and uses of oblique projection. Skills: Produce and interpret simple oblique drawings using appropriate drawing instruments. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can oblique projection communicate the form of an object?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.1 Oblique Projection” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can oblique projection communicate the form of an object?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to observe three-dimensional objects and sample drawings; identify the front face and receding axes; practise drawing basic forms; add visible details and dimensions where required; display and critique drawings using agreed criteria.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can oblique projection communicate the form of an object? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 37

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 10	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.1 Oblique Projection

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Describe the steps for drawing objects using oblique projection. Skills: Produce and interpret simple oblique drawings using appropriate drawing instruments. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can oblique projection communicate the form of an object?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.1 Oblique Projection” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can oblique projection communicate the form of an object?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to observe three-dimensional objects and sample drawings; identify the front face and receding axes; practise drawing basic forms; add visible details and dimensions where required; display and critique drawings using agreed criteria.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can oblique projection communicate the form of an object? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 38

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 10	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.2 Visual Programming

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Explain the meaning and basic elements of visual programming. Skills: Design, test and improve a simple visual programme to solve a given problem. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can visual programming be used to solve everyday problems?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.2 Visual Programming” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can visual programming be used to solve everyday problems?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to explore a suitable visual programming environment or unplugged block cards; identify sequence, events, loops and conditions; plan an algorithm; create a simple programme; test, debug and present the solution.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can visual programming be used to solve everyday problems? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 39

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 10	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.2 Visual Programming

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify visual programming tools and blocks used to create simple programmes. Skills: Design, test and improve a simple visual programme to solve a given problem. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can visual programming be used to solve everyday problems?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.2 Visual Programming” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can visual programming be used to solve everyday problems?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to explore a suitable visual programming environment or unplugged block cards; identify sequence, events, loops and conditions; plan an algorithm; create a simple programme; test, debug and present the solution.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can visual programming be used to solve everyday problems? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 40

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 2	WEEK: 10	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.2 Visual Programming

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Design, test and improve a simple visual programme to solve a given problem. Skills: Design, test and improve a simple visual programme to solve a given problem. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can visual programming be used to solve everyday problems?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.2 Visual Programming” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can visual programming be used to solve everyday problems?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to explore a suitable visual programming environment or unplugged block cards; identify sequence, events, loops and conditions; plan an algorithm; create a simple programme; test, debug and present the solution.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can visual programming be used to solve everyday problems? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 41

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 1	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.2 Visual Programming

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Explain the meaning and basic elements of visual programming. Skills: Design, test and improve a simple visual programme to solve a given problem. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can visual programming be used to solve everyday problems?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.2 Visual Programming” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can visual programming be used to solve everyday problems?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to explore a suitable visual programming environment or unplugged block cards; identify sequence, events, loops and conditions; plan an algorithm; create a simple programme; test, debug and present the solution.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can visual programming be used to solve everyday problems? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 42

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 1	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.2 Visual Programming

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify visual programming tools and blocks used to create simple programmes. Skills: Design, test and improve a simple visual programme to solve a given problem. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can visual programming be used to solve everyday problems?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.2 Visual Programming” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can visual programming be used to solve everyday problems?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to explore a suitable visual programming environment or unplugged block cards; identify sequence, events, loops and conditions; plan an algorithm; create a simple programme; test, debug and present the solution.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can visual programming be used to solve everyday problems? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 43

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 1	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.2 Visual Programming

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Design, test and improve a simple visual programme to solve a given problem. Skills: Design, test and improve a simple visual programme to solve a given problem. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can visual programming be used to solve everyday problems?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.2 Visual Programming” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can visual programming be used to solve everyday problems?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to explore a suitable visual programming environment or unplugged block cards; identify sequence, events, loops and conditions; plan an algorithm; create a simple programme; test, debug and present the solution.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can visual programming be used to solve everyday problems? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 44

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 1	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.2 Visual Programming

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Explain the meaning and basic elements of visual programming. Skills: Design, test and improve a simple visual programme to solve a given problem. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can visual programming be used to solve everyday problems?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.2 Visual Programming” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can visual programming be used to solve everyday problems?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to explore a suitable visual programming environment or unplugged block cards; identify sequence, events, loops and conditions; plan an algorithm; create a simple programme; test, debug and present the solution.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can visual programming be used to solve everyday problems? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 45

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 2	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.2 Visual Programming

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify visual programming tools and blocks used to create simple programmes. Skills: Design, test and improve a simple visual programme to solve a given problem. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can visual programming be used to solve everyday problems?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.2 Visual Programming” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can visual programming be used to solve everyday problems?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to explore a suitable visual programming environment or unplugged block cards; identify sequence, events, loops and conditions; plan an algorithm; create a simple programme; test, debug and present the solution.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can visual programming be used to solve everyday problems? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 46

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 2	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.2 Visual Programming

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Design, test and improve a simple visual programme to solve a given problem. Skills: Design, test and improve a simple visual programme to solve a given problem. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can visual programming be used to solve everyday problems?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.2 Visual Programming” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can visual programming be used to solve everyday problems?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to explore a suitable visual programming environment or unplugged block cards; identify sequence, events, loops and conditions; plan an algorithm; create a simple programme; test, debug and present the solution.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can visual programming be used to solve everyday problems? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 47

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 2	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.2 Visual Programming

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Explain the meaning and basic elements of visual programming. Skills: Design, test and improve a simple visual programme to solve a given problem. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can visual programming be used to solve everyday problems?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.2 Visual Programming” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can visual programming be used to solve everyday problems?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to explore a suitable visual programming environment or unplugged block cards; identify sequence, events, loops and conditions; plan an algorithm; create a simple programme; test, debug and present the solution.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can visual programming be used to solve everyday problems? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 48

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 2	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.2 Visual Programming

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify visual programming tools and blocks used to create simple programmes. Skills: Design, test and improve a simple visual programme to solve a given problem. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can visual programming be used to solve everyday problems?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.2 Visual Programming” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can visual programming be used to solve everyday problems?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to explore a suitable visual programming environment or unplugged block cards; identify sequence, events, loops and conditions; plan an algorithm; create a simple programme; test, debug and present the solution.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can visual programming be used to solve everyday problems? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 49

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 3	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.2 Visual Programming

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Design, test and improve a simple visual programme to solve a given problem. Skills: Design, test and improve a simple visual programme to solve a given problem. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can visual programming be used to solve everyday problems?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.2 Visual Programming” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can visual programming be used to solve everyday problems?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to explore a suitable visual programming environment or unplugged block cards; identify sequence, events, loops and conditions; plan an algorithm; create a simple programme; test, debug and present the solution.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can visual programming be used to solve everyday problems? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 50

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 3	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.2 Visual Programming

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Explain the meaning and basic elements of visual programming. Skills: Design, test and improve a simple visual programme to solve a given problem. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can visual programming be used to solve everyday problems?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.2 Visual Programming” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can visual programming be used to solve everyday problems?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to explore a suitable visual programming environment or unplugged block cards; identify sequence, events, loops and conditions; plan an algorithm; create a simple programme; test, debug and present the solution.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can visual programming be used to solve everyday problems? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 51

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 3	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.2 Visual Programming

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify visual programming tools and blocks used to create simple programmes. Skills: Design, test and improve a simple visual programme to solve a given problem. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can visual programming be used to solve everyday problems?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.2 Visual Programming” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can visual programming be used to solve everyday problems?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to explore a suitable visual programming environment or unplugged block cards; identify sequence, events, loops and conditions; plan an algorithm; create a simple programme; test, debug and present the solution.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can visual programming be used to solve everyday problems? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 52

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 3	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 2.0 Communication in Pre-Technical Studies

Sub-strand: 2.2 Visual Programming

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Design, test and improve a simple visual programme to solve a given problem. Skills: Design, test and improve a simple visual programme to solve a given problem. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can visual programming be used to solve everyday problems?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “2.2 Visual Programming” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can visual programming be used to solve everyday problems?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to explore a suitable visual programming environment or unplugged block cards; identify sequence, events, loops and conditions; plan an algorithm; create a simple programme; test, debug and present the solution.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can visual programming be used to solve everyday problems? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 53

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 4	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 3.0 Materials for Production

Sub-strand: 3.1 Wood

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify common types and properties of wood. Skills: Select and handle suitable wood safely for a simple task. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do the properties of wood influence its use?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “3.1 Wood” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do the properties of wood influence its use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to observe wood samples or images; compare grain, texture and other properties; classify examples; relate properties to uses; discuss sustainable sourcing and practise safe handling of samples.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do the properties of wood influence its use? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 54

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 4	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 3.0 Materials for Production

Sub-strand: 3.1 Wood

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Describe uses of different types of wood in production. Skills: Select and handle suitable wood safely for a simple task. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do the properties of wood influence its use?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “3.1 Wood” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do the properties of wood influence its use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to observe wood samples or images; compare grain, texture and other properties; classify examples; relate properties to uses; discuss sustainable sourcing and practise safe handling of samples.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do the properties of wood influence its use? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 55

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 4	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 3.0 Materials for Production

Sub-strand: 3.1 Wood

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Select and handle suitable wood safely for a simple task. Skills: Select and handle suitable wood safely for a simple task. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do the properties of wood influence its use?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “3.1 Wood” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do the properties of wood influence its use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to observe wood samples or images; compare grain, texture and other properties; classify examples; relate properties to uses; discuss sustainable sourcing and practise safe handling of samples.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do the properties of wood influence its use? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 56

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 4	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 3.0 Materials for Production

Sub-strand: 3.1 Wood

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify common types and properties of wood. Skills: Select and handle suitable wood safely for a simple task. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do the properties of wood influence its use?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “3.1 Wood” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do the properties of wood influence its use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to observe wood samples or images; compare grain, texture and other properties; classify examples; relate properties to uses; discuss sustainable sourcing and practise safe handling of samples.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do the properties of wood influence its use? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 57

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 5	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 3.0 Materials for Production

Sub-strand: 3.1 Wood

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Describe uses of different types of wood in production. Skills: Select and handle suitable wood safely for a simple task. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do the properties of wood influence its use?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “3.1 Wood” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do the properties of wood influence its use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to observe wood samples or images; compare grain, texture and other properties; classify examples; relate properties to uses; discuss sustainable sourcing and practise safe handling of samples.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do the properties of wood influence its use? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 58

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 5	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 3.0 Materials for Production

Sub-strand: 3.1 Wood

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Select and handle suitable wood safely for a simple task. Skills: Select and handle suitable wood safely for a simple task. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do the properties of wood influence its use?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “3.1 Wood” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do the properties of wood influence its use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to observe wood samples or images; compare grain, texture and other properties; classify examples; relate properties to uses; discuss sustainable sourcing and practise safe handling of samples.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do the properties of wood influence its use? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 59

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 5	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 3.0 Materials for Production

Sub-strand: 3.1 Wood

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify common types and properties of wood. Skills: Select and handle suitable wood safely for a simple task. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do the properties of wood influence its use?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “3.1 Wood” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do the properties of wood influence its use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to observe wood samples or images; compare grain, texture and other properties; classify examples; relate properties to uses; discuss sustainable sourcing and practise safe handling of samples.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do the properties of wood influence its use? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 60

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 5	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 3.0 Materials for Production

Sub-strand: 3.1 Wood

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Describe uses of different types of wood in production. Skills: Select and handle suitable wood safely for a simple task. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do the properties of wood influence its use?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “3.1 Wood” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do the properties of wood influence its use?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to observe wood samples or images; compare grain, texture and other properties; classify examples; relate properties to uses; discuss sustainable sourcing and practise safe handling of samples.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do the properties of wood influence its use? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 61

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 6	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 3.0 Materials for Production

Sub-strand: 3.2 Handling Waste Materials

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify waste materials generated in homes, school and workplaces. Skills: Develop a useful product or plan from selected safe waste materials. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can waste materials be managed to support production and environmental care?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “3.2 Handling Waste Materials” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can waste materials be managed to support production and environmental care?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to conduct a waste audit in a safe area; sort clean materials; discuss the 3Rs and environmental effects; design and make a simple useful item from suitable materials; present the product and explain its value.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can waste materials be managed to support production and environmental care? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 62

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 6	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 3.0 Materials for Production

Sub-strand: 3.2 Handling Waste Materials

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Explain ways of sorting, reusing, recycling and safely disposing of waste materials. Skills: Develop a useful product or plan from selected safe waste materials. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can waste materials be managed to support production and environmental care?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “3.2 Handling Waste Materials” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can waste materials be managed to support production and environmental care?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to conduct a waste audit in a safe area; sort clean materials; discuss the 3Rs and environmental effects; design and make a simple useful item from suitable materials; present the product and explain its value.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can waste materials be managed to support production and environmental care? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 63

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 6	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 3.0 Materials for Production

Sub-strand: 3.2 Handling Waste Materials

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Develop a useful product or plan from selected safe waste materials. Skills: Develop a useful product or plan from selected safe waste materials. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can waste materials be managed to support production and environmental care?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “3.2 Handling Waste Materials” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can waste materials be managed to support production and environmental care?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to conduct a waste audit in a safe area; sort clean materials; discuss the 3Rs and environmental effects; design and make a simple useful item from suitable materials; present the product and explain its value.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can waste materials be managed to support production and environmental care? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 64

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 6	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 3.0 Materials for Production

Sub-strand: 3.2 Handling Waste Materials

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify waste materials generated in homes, school and workplaces. Skills: Develop a useful product or plan from selected safe waste materials. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can waste materials be managed to support production and environmental care?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “3.2 Handling Waste Materials” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can waste materials be managed to support production and environmental care?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to conduct a waste audit in a safe area; sort clean materials; discuss the 3Rs and environmental effects; design and make a simple useful item from suitable materials; present the product and explain its value.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can waste materials be managed to support production and environmental care? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 65

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 7	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 3.0 Materials for Production

Sub-strand: 3.2 Handling Waste Materials

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Explain ways of sorting, reusing, recycling and safely disposing of waste materials. Skills: Develop a useful product or plan from selected safe waste materials. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can waste materials be managed to support production and environmental care?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “3.2 Handling Waste Materials” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can waste materials be managed to support production and environmental care?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to conduct a waste audit in a safe area; sort clean materials; discuss the 3Rs and environmental effects; design and make a simple useful item from suitable materials; present the product and explain its value.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can waste materials be managed to support production and environmental care? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 66

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 7	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 3.0 Materials for Production

Sub-strand: 3.2 Handling Waste Materials

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Develop a useful product or plan from selected safe waste materials. Skills: Develop a useful product or plan from selected safe waste materials. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can waste materials be managed to support production and environmental care?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “3.2 Handling Waste Materials” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can waste materials be managed to support production and environmental care?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to conduct a waste audit in a safe area; sort clean materials; discuss the 3Rs and environmental effects; design and make a simple useful item from suitable materials; present the product and explain its value.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can waste materials be managed to support production and environmental care? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 67

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 7	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 3.0 Materials for Production

Sub-strand: 3.2 Handling Waste Materials

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify waste materials generated in homes, school and workplaces. Skills: Develop a useful product or plan from selected safe waste materials. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can waste materials be managed to support production and environmental care?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “3.2 Handling Waste Materials” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can waste materials be managed to support production and environmental care?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to conduct a waste audit in a safe area; sort clean materials; discuss the 3Rs and environmental effects; design and make a simple useful item from suitable materials; present the product and explain its value.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can waste materials be managed to support production and environmental care? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 68

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 7	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 3.0 Materials for Production

Sub-strand: 3.2 Handling Waste Materials

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Explain ways of sorting, reusing, recycling and safely disposing of waste materials. Skills: Develop a useful product or plan from selected safe waste materials. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can waste materials be managed to support production and environmental care?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “3.2 Handling Waste Materials” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can waste materials be managed to support production and environmental care?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to conduct a waste audit in a safe area; sort clean materials; discuss the 3Rs and environmental effects; design and make a simple useful item from suitable materials; present the product and explain its value.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can waste materials be managed to support production and environmental care? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 69

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 8	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.1 Holding Tools

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify common holding tools and their functions. Skills: Demonstrate safe use, care and storage of holding tools. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do holding tools support safe and accurate work?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.1 Holding Tools” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do holding tools support safe and accurate work?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to identify pliers, clamps, vices, tongs and clips from realia or pictures; match tools to tasks; practise securing materials under supervision; inspect, clean and store tools safely.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do holding tools support safe and accurate work? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 70

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 8	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.1 Holding Tools

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Select appropriate holding tools for given tasks. Skills: Demonstrate safe use, care and storage of holding tools. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do holding tools support safe and accurate work?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.1 Holding Tools” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do holding tools support safe and accurate work?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to identify pliers, clamps, vices, tongs and clips from realia or pictures; match tools to tasks; practise securing materials under supervision; inspect, clean and store tools safely.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do holding tools support safe and accurate work? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 71

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 8	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.1 Holding Tools

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Demonstrate safe use, care and storage of holding tools. Skills: Demonstrate safe use, care and storage of holding tools. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do holding tools support safe and accurate work?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.1 Holding Tools” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do holding tools support safe and accurate work?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to identify pliers, clamps, vices, tongs and clips from realia or pictures; match tools to tasks; practise securing materials under supervision; inspect, clean and store tools safely.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do holding tools support safe and accurate work? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 72

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 8	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.1 Holding Tools

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify common holding tools and their functions. Skills: Demonstrate safe use, care and storage of holding tools. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do holding tools support safe and accurate work?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.1 Holding Tools” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do holding tools support safe and accurate work?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to identify pliers, clamps, vices, tongs and clips from realia or pictures; match tools to tasks; practise securing materials under supervision; inspect, clean and store tools safely.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do holding tools support safe and accurate work? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 73

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 9	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.1 Holding Tools

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Select appropriate holding tools for given tasks. Skills: Demonstrate safe use, care and storage of holding tools. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do holding tools support safe and accurate work?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.1 Holding Tools” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do holding tools support safe and accurate work?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to identify pliers, clamps, vices, tongs and clips from realia or pictures; match tools to tasks; practise securing materials under supervision; inspect, clean and store tools safely.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do holding tools support safe and accurate work? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 74

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 9	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.1 Holding Tools

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Demonstrate safe use, care and storage of holding tools. Skills: Demonstrate safe use, care and storage of holding tools. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do holding tools support safe and accurate work?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.1 Holding Tools” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do holding tools support safe and accurate work?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to identify pliers, clamps, vices, tongs and clips from realia or pictures; match tools to tasks; practise securing materials under supervision; inspect, clean and store tools safely.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do holding tools support safe and accurate work? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 75

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 9	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.1 Holding Tools

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify common holding tools and their functions. Skills: Demonstrate safe use, care and storage of holding tools. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do holding tools support safe and accurate work?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.1 Holding Tools” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do holding tools support safe and accurate work?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to identify pliers, clamps, vices, tongs and clips from realia or pictures; match tools to tasks; practise securing materials under supervision; inspect, clean and store tools safely.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do holding tools support safe and accurate work? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 76

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 9	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.1 Holding Tools

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Select appropriate holding tools for given tasks. Skills: Demonstrate safe use, care and storage of holding tools. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do holding tools support safe and accurate work?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.1 Holding Tools” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do holding tools support safe and accurate work?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to identify pliers, clamps, vices, tongs and clips from realia or pictures; match tools to tasks; practise securing materials under supervision; inspect, clean and store tools safely.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do holding tools support safe and accurate work? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 77

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 10	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.2 Driving Tools

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify common driving tools and their functions. Skills: Demonstrate safe use, care and storage of driving tools. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How are driving tools used in day-to-day production tasks?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.2 Driving Tools” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How are driving tools used in day-to-day production tasks?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to identify hammers, screwdrivers, spanners, punches and mallets; match tools to fasteners and tasks; practise safe handling on suitable materials under supervision; clean and store tools correctly.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How are driving tools used in day-to-day production tasks? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 78

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 10	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.2 Driving Tools

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Select suitable driving tools for given tasks. Skills: Demonstrate safe use, care and storage of driving tools. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How are driving tools used in day-to-day production tasks?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.2 Driving Tools” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How are driving tools used in day-to-day production tasks?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to identify hammers, screwdrivers, spanners, punches and mallets; match tools to fasteners and tasks; practise safe handling on suitable materials under supervision; clean and store tools correctly.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How are driving tools used in day-to-day production tasks? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 79

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 10	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.2 Driving Tools

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Demonstrate safe use, care and storage of driving tools. Skills: Demonstrate safe use, care and storage of driving tools. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How are driving tools used in day-to-day production tasks?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.2 Driving Tools” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How are driving tools used in day-to-day production tasks?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to identify hammers, screwdrivers, spanners, punches and mallets; match tools to fasteners and tasks; practise safe handling on suitable materials under supervision; clean and store tools correctly.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How are driving tools used in day-to-day production tasks? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 80

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 3	WEEK: 10	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.2 Driving Tools

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify common driving tools and their functions. Skills: Demonstrate safe use, care and storage of driving tools. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How are driving tools used in day-to-day production tasks?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.2 Driving Tools” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How are driving tools used in day-to-day production tasks?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to identify hammers, screwdrivers, spanners, punches and mallets; match tools to fasteners and tasks; practise safe handling on suitable materials under supervision; clean and store tools correctly.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How are driving tools used in day-to-day production tasks? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 81

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 1	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.2 Driving Tools

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Select suitable driving tools for given tasks. Skills: Demonstrate safe use, care and storage of driving tools. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How are driving tools used in day-to-day production tasks?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.2 Driving Tools” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How are driving tools used in day-to-day production tasks?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to identify hammers, screwdrivers, spanners, punches and mallets; match tools to fasteners and tasks; practise safe handling on suitable materials under supervision; clean and store tools correctly.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How are driving tools used in day-to-day production tasks? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 82

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 1	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.2 Driving Tools

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Demonstrate safe use, care and storage of driving tools. Skills: Demonstrate safe use, care and storage of driving tools. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How are driving tools used in day-to-day production tasks?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.2 Driving Tools” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How are driving tools used in day-to-day production tasks?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to identify hammers, screwdrivers, spanners, punches and mallets; match tools to fasteners and tasks; practise safe handling on suitable materials under supervision; clean and store tools correctly.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How are driving tools used in day-to-day production tasks? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 83

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 1	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.2 Driving Tools

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify common driving tools and their functions. Skills: Demonstrate safe use, care and storage of driving tools. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How are driving tools used in day-to-day production tasks?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.2 Driving Tools” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How are driving tools used in day-to-day production tasks?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to identify hammers, screwdrivers, spanners, punches and mallets; match tools to fasteners and tasks; practise safe handling on suitable materials under supervision; clean and store tools correctly.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How are driving tools used in day-to-day production tasks? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 84

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 1	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.2 Driving Tools

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Select suitable driving tools for given tasks. Skills: Demonstrate safe use, care and storage of driving tools. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How are driving tools used in day-to-day production tasks?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.2 Driving Tools” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How are driving tools used in day-to-day production tasks?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to identify hammers, screwdrivers, spanners, punches and mallets; match tools to fasteners and tasks; practise safe handling on suitable materials under supervision; clean and store tools correctly.

Step 1: Learners observe and classify materials, tools or finished products by visible properties, use and suitability.

Step 2: Learners discuss the sequence of work and select appropriate materials, tools and protective equipment for the task.

Step 3: Under supervision, learners demonstrate or simulate the relevant making, joining, shaping, finishing or maintenance process using safe materials.

Step 4: Learners inspect the product or process, record strengths and areas for improvement, and tidy the work area safely.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How are driving tools used in day-to-day production tasks? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 85

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 2	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify a relevant problem in the school or community that can be addressed through a practical project. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 86

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 2	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Develop a plan showing the design, materials, tools, processes, cost and safety measures for the project. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 87

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 2	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 88

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 2	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify a relevant problem in the school or community that can be addressed through a practical project. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 89

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 3	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Develop a plan showing the design, materials, tools, processes, cost and safety measures for the project. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 90

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 3	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 91

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 3	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify a relevant problem in the school or community that can be addressed through a practical project. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 92

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 3	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Develop a plan showing the design, materials, tools, processes, cost and safety measures for the project. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 93

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 4	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 94

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 4	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify a relevant problem in the school or community that can be addressed through a practical project. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 95

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 4	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Develop a plan showing the design, materials, tools, processes, cost and safety measures for the project. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 96

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 4	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 97

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 5	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify a relevant problem in the school or community that can be addressed through a practical project. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 98

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 5	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Develop a plan showing the design, materials, tools, processes, cost and safety measures for the project. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 99

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 5	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 100

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 5	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify a relevant problem in the school or community that can be addressed through a practical project. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 101

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 6	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Develop a plan showing the design, materials, tools, processes, cost and safety measures for the project. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 102

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 6	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 103

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 6	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify a relevant problem in the school or community that can be addressed through a practical project. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 104

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 6	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 4.0 Tools and Production

Sub-strand: 4.3 Project

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Develop a plan showing the design, materials, tools, processes, cost and safety measures for the project. Skills: Produce, evaluate and present a functional project while demonstrating teamwork and responsible use of resources. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How can a practical project address a need in the community?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “4.3 Project” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How can a practical project address a need in the community?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to identify and define a manageable need; consult intended users; sketch and refine a design; prepare a materials and tool list, cost estimate and work plan; make the product safely; test it against criteria; improve and present the completed project.

Step 1: Learners observe sample drawings and identify drawing conventions, symbols, dimensions or communication features.

Step 2: Learners select suitable instruments and plan the drawing or design task, observing correct handling and care of equipment.

Step 3: Learners produce a labelled sketch, drawing, model or design using appropriate procedures and accurate measurements where required.

Step 4: Learners compare work with agreed criteria, make corrections and explain how the drawing communicates the intended idea.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How can a practical project address a need in the community? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 105

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 7	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 5.0 Entrepreneurship

Sub-strand: 5.1 Financial Services

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify financial services and institutions available in the community. Skills: Compare suitable financial services for a given need while considering safety and responsibility. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do financial services support individuals and businesses?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “5.1 Financial Services” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do financial services support individuals and businesses?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to identify banks, SACCOs, mobile money and other financial service providers; examine age-appropriate examples of services; discuss saving, payments and borrowing; identify safe practices for protecting financial information.

Step 1: Learners examine examples, diagrams, tools or a practical problem and identify important features.

Step 2: Learners discuss the relevant principles and plan a suitable solution in pairs or groups.

Step 3: Learners carry out a guided design, measurement, demonstration or problem-solving task using appropriate resources.

Step 4: Learners present the outcome, evaluate it against simple criteria and reflect on safe and responsible practice.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do financial services support individuals and businesses? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 106

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 7	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 5.0 Entrepreneurship

Sub-strand: 5.1 Financial Services

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Explain the importance of financial services in personal and business activities. Skills: Compare suitable financial services for a given need while considering safety and responsibility. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do financial services support individuals and businesses?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect "5.1 Financial Services" to learners' experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do financial services support individuals and businesses?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to identify banks, SACCOs, mobile money and other financial service providers; examine age-appropriate examples of services; discuss saving, payments and borrowing; identify safe practices for protecting financial information.

Step 1: Learners examine examples, diagrams, tools or a practical problem and identify important features.

Step 2: Learners discuss the relevant principles and plan a suitable solution in pairs or groups.

Step 3: Learners carry out a guided design, measurement, demonstration or problem-solving task using appropriate resources.

Step 4: Learners present the outcome, evaluate it against simple criteria and reflect on safe and responsible practice.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do financial services support individuals and businesses? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 107

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 7	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 5.0 Entrepreneurship

Sub-strand: 5.1 Financial Services

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Compare suitable financial services for a given need while considering safety and responsibility. Skills: Compare suitable financial services for a given need while considering safety and responsibility. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do financial services support individuals and businesses?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “5.1 Financial Services” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do financial services support individuals and businesses?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to identify banks, SACCOs, mobile money and other financial service providers; examine age-appropriate examples of services; discuss saving, payments and borrowing; identify safe practices for protecting financial information.

Step 1: Learners examine examples, diagrams, tools or a practical problem and identify important features.

Step 2: Learners discuss the relevant principles and plan a suitable solution in pairs or groups.

Step 3: Learners carry out a guided design, measurement, demonstration or problem-solving task using appropriate resources.

Step 4: Learners present the outcome, evaluate it against simple criteria and reflect on safe and responsible practice.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do financial services support individuals and businesses? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 108

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 7	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 5.0 Entrepreneurship

Sub-strand: 5.1 Financial Services

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify financial services and institutions available in the community. Skills: Compare suitable financial services for a given need while considering safety and responsibility. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How do financial services support individuals and businesses?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “5.1 Financial Services” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How do financial services support individuals and businesses?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to identify banks, SACCOs, mobile money and other financial service providers; examine age-appropriate examples of services; discuss saving, payments and borrowing; identify safe practices for protecting financial information.

Step 1: Learners examine examples, diagrams, tools or a practical problem and identify important features.

Step 2: Learners discuss the relevant principles and plan a suitable solution in pairs or groups.

Step 3: Learners carry out a guided design, measurement, demonstration or problem-solving task using appropriate resources.

Step 4: Learners present the outcome, evaluate it against simple criteria and reflect on safe and responsible practice.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How do financial services support individuals and businesses? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 109

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 8	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 5.0 Entrepreneurship

Sub-strand: 5.2 Government and Business

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify ways in which government supports and regulates businesses. Skills: Analyse how government-business relationships can promote responsible enterprise. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How does government interact with businesses?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “5.2 Government and Business” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How does government interact with businesses?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to brainstorm local businesses; research or discuss relevant public services, licensing, taxation and consumer protection at an age-appropriate level; role-play a responsible business interaction; present findings.

Step 1: Learners examine examples, diagrams, tools or a practical problem and identify important features.

Step 2: Learners discuss the relevant principles and plan a suitable solution in pairs or groups.

Step 3: Learners carry out a guided design, measurement, demonstration or problem-solving task using appropriate resources.

Step 4: Learners present the outcome, evaluate it against simple criteria and reflect on safe and responsible practice.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How does government interact with businesses? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 110

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 8	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 5.0 Entrepreneurship

Sub-strand: 5.2 Government and Business

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Explain selected responsibilities of businesses to government and the community. Skills: Analyse how government-business relationships can promote responsible enterprise. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How does government interact with businesses?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “5.2 Government and Business” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How does government interact with businesses?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to brainstorm local businesses; research or discuss relevant public services, licensing, taxation and consumer protection at an age-appropriate level; role-play a responsible business interaction; present findings.

Step 1: Learners examine examples, diagrams, tools or a practical problem and identify important features.

Step 2: Learners discuss the relevant principles and plan a suitable solution in pairs or groups.

Step 3: Learners carry out a guided design, measurement, demonstration or problem-solving task using appropriate resources.

Step 4: Learners present the outcome, evaluate it against simple criteria and reflect on safe and responsible practice.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How does government interact with businesses? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 111

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 8	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 5.0 Entrepreneurship

Sub-strand: 5.2 Government and Business

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Analyse how government-business relationships can promote responsible enterprise. Skills: Analyse how government-business relationships can promote responsible enterprise. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How does government interact with businesses?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “5.2 Government and Business” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How does government interact with businesses?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to brainstorm local businesses; research or discuss relevant public services, licensing, taxation and consumer protection at an age-appropriate level; role-play a responsible business interaction; present findings.

Step 1: Learners examine examples, diagrams, tools or a practical problem and identify important features.

Step 2: Learners discuss the relevant principles and plan a suitable solution in pairs or groups.

Step 3: Learners carry out a guided design, measurement, demonstration or problem-solving task using appropriate resources.

Step 4: Learners present the outcome, evaluate it against simple criteria and reflect on safe and responsible practice.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How does government interact with businesses? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 112

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 8	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 5.0 Entrepreneurship

Sub-strand: 5.2 Government and Business

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify ways in which government supports and regulates businesses. Skills: Analyse how government-business relationships can promote responsible enterprise. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How does government interact with businesses?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “5.2 Government and Business” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How does government interact with businesses?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to brainstorm local businesses; research or discuss relevant public services, licensing, taxation and consumer protection at an age-appropriate level; role-play a responsible business interaction; present findings.

Step 1: Learners examine examples, diagrams, tools or a practical problem and identify important features.

Step 2: Learners discuss the relevant principles and plan a suitable solution in pairs or groups.

Step 3: Learners carry out a guided design, measurement, demonstration or problem-solving task using appropriate resources.

Step 4: Learners present the outcome, evaluate it against simple criteria and reflect on safe and responsible practice.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How does government interact with businesses? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 113

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 9	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 5.0 Entrepreneurship

Sub-strand: 5.2 Government and Business

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Explain selected responsibilities of businesses to government and the community. Skills: Analyse how government-business relationships can promote responsible enterprise. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How does government interact with businesses?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “5.2 Government and Business” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How does government interact with businesses?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to brainstorm local businesses; research or discuss relevant public services, licensing, taxation and consumer protection at an age-appropriate level; role-play a responsible business interaction; present findings.

Step 1: Learners examine examples, diagrams, tools or a practical problem and identify important features.

Step 2: Learners discuss the relevant principles and plan a suitable solution in pairs or groups.

Step 3: Learners carry out a guided design, measurement, demonstration or problem-solving task using appropriate resources.

Step 4: Learners present the outcome, evaluate it against simple criteria and reflect on safe and responsible practice.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How does government interact with businesses? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 114

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 9	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 5.0 Entrepreneurship

Sub-strand: 5.2 Government and Business

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Analyse how government-business relationships can promote responsible enterprise. Skills: Analyse how government-business relationships can promote responsible enterprise. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: How does government interact with businesses?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “5.2 Government and Business” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: How does government interact with businesses?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to brainstorm local businesses; research or discuss relevant public services, licensing, taxation and consumer protection at an age-appropriate level; role-play a responsible business interaction; present findings.

Step 1: Learners examine examples, diagrams, tools or a practical problem and identify important features.

Step 2: Learners discuss the relevant principles and plan a suitable solution in pairs or groups.

Step 3: Learners carry out a guided design, measurement, demonstration or problem-solving task using appropriate resources.

Step 4: Learners present the outcome, evaluate it against simple criteria and reflect on safe and responsible practice.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: How does government interact with businesses? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 115

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 9	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 5.0 Entrepreneurship

Sub-strand: 5.3 Business Plan

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify the main components of a simple business plan. Skills: Prepare and present a simple business plan for a realistic product or service. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: Why is a business plan important when starting an enterprise?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “5.3 Business Plan” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: Why is a business plan important when starting an enterprise?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to identify a simple business opportunity; describe a product or service and target customers; estimate start-up needs and costs; plan marketing and operations; prepare and pitch a simple business plan.

Step 1: Learners examine examples, diagrams, tools or a practical problem and identify important features.

Step 2: Learners discuss the relevant principles and plan a suitable solution in pairs or groups.

Step 3: Learners carry out a guided design, measurement, demonstration or problem-solving task using appropriate resources.

Step 4: Learners present the outcome, evaluate it against simple criteria and reflect on safe and responsible practice.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: Why is a business plan important when starting an enterprise? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 116

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 9	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 5.0 Entrepreneurship

Sub-strand: 5.3 Business Plan

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Explain the importance of planning before starting a business. Skills: Prepare and present a simple business plan for a realistic product or service. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: Why is a business plan important when starting an enterprise?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “5.3 Business Plan” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: Why is a business plan important when starting an enterprise?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to identify a simple business opportunity; describe a product or service and target customers; estimate start-up needs and costs; plan marketing and operations; prepare and pitch a simple business plan.

Step 1: Learners examine examples, diagrams, tools or a practical problem and identify important features.

Step 2: Learners discuss the relevant principles and plan a suitable solution in pairs or groups.

Step 3: Learners carry out a guided design, measurement, demonstration or problem-solving task using appropriate resources.

Step 4: Learners present the outcome, evaluate it against simple criteria and reflect on safe and responsible practice.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: Why is a business plan important when starting an enterprise? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 117

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 10	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: 5.0 Entrepreneurship

Sub-strand: 5.3 Business Plan

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Prepare and present a simple business plan for a realistic product or service. Skills: Prepare and present a simple business plan for a realistic product or service. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: Why is a business plan important when starting an enterprise?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “5.3 Business Plan” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: Why is a business plan important when starting an enterprise?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to investigate the concept through a guided demonstration or practical activity; steps and observations are recorded. The learner is guided to identify a simple business opportunity; describe a product or service and target customers; estimate start-up needs and costs; plan marketing and operations; prepare and pitch a simple business plan.

Step 1: Learners examine examples, diagrams, tools or a practical problem and identify important features.

Step 2: Learners discuss the relevant principles and plan a suitable solution in pairs or groups.

Step 3: Learners carry out a guided design, measurement, demonstration or problem-solving task using appropriate resources.

Step 4: Learners present the outcome, evaluate it against simple criteria and reflect on safe and responsible practice.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: Why is a business plan important when starting an enterprise? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 118

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 10	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: 5.0 Entrepreneurship

Sub-strand: 5.3 Business Plan

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Identify the main components of a simple business plan. Skills: Prepare and present a simple business plan for a realistic product or service. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: Why is a business plan important when starting an enterprise?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “5.3 Business Plan” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: Why is a business plan important when starting an enterprise?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to apply the skill in a task, evaluate the outcome using agreed criteria, give peer feedback and suggest improvements. The learner is guided to identify a simple business opportunity; describe a product or service and target customers; estimate start-up needs and costs; plan marketing and operations; prepare and pitch a simple business plan.

Step 1: Learners examine examples, diagrams, tools or a practical problem and identify important features.

Step 2: Learners discuss the relevant principles and plan a suitable solution in pairs or groups.

Step 3: Learners carry out a guided design, measurement, demonstration or problem-solving task using appropriate resources.

Step 4: Learners present the outcome, evaluate it against simple criteria and reflect on safe and responsible practice.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: Why is a business plan important when starting an enterprise? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 119

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 10	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: 5.0 Entrepreneurship

Sub-strand: 5.3 Business Plan

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Explain the importance of planning before starting a business. Skills: Prepare and present a simple business plan for a realistic product or service. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: Why is a business plan important when starting an enterprise?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “5.3 Business Plan” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: Why is a business plan important when starting an enterprise?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to observe examples, identify features and discuss prior knowledge in pairs; findings are recorded in a learning journal. The learner is guided to identify a simple business opportunity; describe a product or service and target customers; estimate start-up needs and costs; plan marketing and operations; prepare and pitch a simple business plan.

Step 1: Learners examine examples, diagrams, tools or a practical problem and identify important features.

Step 2: Learners discuss the relevant principles and plan a suitable solution in pairs or groups.

Step 3: Learners carry out a guided design, measurement, demonstration or problem-solving task using appropriate resources.

Step 4: Learners present the outcome, evaluate it against simple criteria and reflect on safe and responsible practice.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: Why is a business plan important when starting an enterprise? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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GRADE 9 PRE-TECHNICAL STUDIES – LESSON PLAN 120

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Pre-Technical Studies	DATE: _____
TERM: 4	WEEK: 10	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: 5.0 Entrepreneurship

Sub-strand: 5.3 Business Plan

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** Prepare and present a simple business plan for a realistic product or service. Skills: Prepare and present a simple business plan for a realistic product or service. Attitude: demonstrate responsibility, cooperation, creativity and commitment to safe and ethical practices.

Key Inquiry Question: Why is a business plan important when starting an enterprise?

Learning Resources: Relevant tools/equipment and safe materials; PPE where appropriate; charts, drawings or digital devices; learner's book; teacher-prepared worksheets.

Organization of Learning Learners work individually, in pairs and in groups. Practical activities use appropriate tools, materials and personal protective equipment under supervision. Learners follow workshop safety rules, keep the work area orderly and report hazards immediately.

Lesson Development

5 minutes – Introduction

Connect “5.3 Business Plan” to learners’ experiences at home, school or in the community. Present a safe example, drawing, model, tool or design problem. Learners observe, share prior knowledge and consider the inquiry question: Why is a business plan important when starting an enterprise?

30 minutes – Lesson Development

Learning experiences from the scheme: The learner is guided to read instructions, interpret diagrams and compare examples; groups classify information and present conclusions. The learner is guided to identify a simple business opportunity; describe a product or service and target customers; estimate start-up needs and costs; plan marketing and operations; prepare and pitch a simple business plan.

Step 1: Learners examine examples, diagrams, tools or a practical problem and identify important features.

Step 2: Learners discuss the relevant principles and plan a suitable solution in pairs or groups.

Step 3: Learners carry out a guided design, measurement, demonstration or problem-solving task using appropriate resources.

Step 4: Learners present the outcome, evaluate it against simple criteria and reflect on safe and responsible practice.

Extended Activities

Learners who complete the task extend their work by improving a design, preparing a labelled diagram, comparing alternative materials or identifying a real-life application. Learners needing support use step-by-step picture guides, templates, peer support and supervised demonstrations. Do not use hazardous machinery, electrical systems, heights, sharp tools or chemicals without approved facilities, trained supervision and required protective equipment.

5 minutes – Conclusion

Learners summarise the main concept or safe procedure and answer the inquiry question: Why is a business plan important when starting an enterprise? They complete an exit task by identifying a key feature, explaining a safety measure or evaluating one aspect of their design or practical work.

Assessment

Suggested assessment evidence: Oral questions, observation checklist, written task, practical performance, portfolio and peer/self-assessment as appropriate.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence of technical knowledge and practical skills, safe use of tools/materials, challenges encountered, learners requiring support and the next instructional step.

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Implementation Notes

This collection contains 120 Grade 9 Pre-Technical Studies lesson plans organised across Terms 1, 2 and 3 according to the teacher-prepared annual scheme. The scheme lists 120 suggested lessons across 13 sub-strands.

The allocation of 40 lessons per term and 4 lessons per week over 10 teaching weeks is a planning proposal, not a KICD-prescribed term allocation. Adjust pacing to the school calendar, learner needs, availability of tools/materials and assessment requirements.

This is an original teacher-prepared resource aligned to the Grade 9 Pre-Technical Studies curriculum design, not an official KICD publication. Practical activities require appropriate facilities, supervision, safe materials and protective equipment.